

WARBURTO POINT, Australia

WMO: 946660

Lat: 34.004S

Lon: 137.528E

Elev: 1

StdP: 101.31

Time Zone: 9.50 (AUA)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.9	6.9	-3.6	2.8	23.8	-1.8	3.3	21.6	12.3	14.9	11.5	15.2	1.1	0	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.0	38.3	19.7	36.2	19.3	34.0	18.8	22.7	30.8	21.9	29.6	21.1	28.6	3.8	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.4	15.1	25.8	19.5	14.2	25.3	18.7	13.5	24.9	67.1	31.2	63.9	29.4	61.2	28.6	34.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.3	11.9	10.5	3.9	42.8	0.9	1.7	3.3	44.0	2.7	45.0	2.2	46.0	1.6	47.2		
			2.8	25.7	1.0	3.6	2.1	28.2	1.5	30.3	0.9	32.3	0.2	34.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.2	25.3	25.6	23.7	19.9	16.1	13.0	12.3	13.7	16.1	19.0	22.2	24.0
	DBStd	5.77	4.23	3.89	3.84	3.28	2.58	2.19	1.91	2.54	3.28	3.37	4.02	4.00
	HDD10.0	4	0	0	0	0	0	2	2	1	0	0	0	0
	HDD18.3	716	0	0	1	17	79	161	187	146	85	32	7	1
	CDD10.0	3364	474	438	424	297	190	91	73	117	184	278	364	434
	CDD18.3	1034	216	205	167	64	11	1	0	3	18	51	121	177
	CDH23.3	9109	2133	1896	1331	377	38	2	1	17	156	457	1062	1639
CDH26.7	4194	1103	922	581	123	2	0	0	2	42	169	472	777	
Wind	WSAvg	4.1	5.5	5.2	4.0	2.7	2.4	2.4	2.7	3.6	4.3	5.2	5.5	5.8
Precipitation	PrecAvg	351	14	18	15	27	41	45	45	40	41	31	21	17
	PrecMax	544	68	66	50	137	100	123	91	76	117	97	47	53
	PrecMin	211	0	0	0	1	4	2	11	7	9	3	0	0
	PrecStd	77	15	20	14	26	23	25	18	17	24	23	13	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.8	40.9	38.9	33.4	26.7	22.0	21.3	25.7	30.4	34.6	38.5	38.8
		MCWB	20.1	22.3	19.8	17.7	15.4	13.5	12.1	14.3	15.1	15.8	18.4	19.0
	2%	DB	38.1	37.5	35.2	30.1	24.2	19.6	19.0	22.5	27.4	30.9	35.0	36.2
		MCWB	20.0	20.4	19.7	16.4	15.2	12.9	11.5	12.2	14.3	15.0	17.7	18.8
	5%	DB	35.5	34.5	32.5	27.4	22.2	18.0	17.5	20.0	24.6	28.2	32.0	33.6
		MCWB	19.5	20.1	18.8	16.0	14.3	12.4	11.0	11.8	13.2	14.5	17.2	18.5
	10%	DB	32.5	32.1	29.8	25.1	20.5	16.7	16.1	18.1	22.1	25.8	29.4	31.0
		MCWB	19.2	19.5	18.7	15.9	13.8	11.7	10.7	11.2	12.7	14.3	17.0	18.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.7	27.2	23.2	20.1	17.1	15.3	14.5	15.5	17.1	18.2	21.7	22.9
		MCDB	32.2	36.7	30.3	26.4	22.1	19.2	16.7	21.8	25.5	24.5	31.3	31.4
	2%	WB	22.4	22.9	21.9	18.7	16.2	14.2	12.8	13.6	15.7	17.2	20.4	21.7
		MCDB	30.6	30.8	29.2	24.1	21.2	17.8	16.7	18.9	23.3	24.0	28.3	29.3
	5%	WB	21.5	21.7	20.9	17.7	15.6	13.3	12.0	12.7	14.5	16.4	19.4	20.8
		MCDB	29.8	29.4	26.9	23.8	20.4	16.4	15.9	17.7	21.6	23.3	26.8	27.9
	10%	WB	20.6	20.9	20.0	16.9	14.8	12.6	11.3	11.9	13.5	15.6	18.5	20.0
		MCDB	29.2	28.3	26.4	23.0	19.4	15.5	15.0	16.7	20.1	22.6	25.7	27.2
Mean Daily Temperature Range	5% DB	MDBR	10.4	10.0	9.5	8.8	7.8	7.2	7.6	8.2	9.3	10.0	10.0	10.0
		MCDBR	14.7	13.9	13.4	12.3	10.3	8.7	9.8	11.7	13.6	15.5	14.1	13.8
		MCWBR	4.6	4.7	3.8	4.4	4.4	4.3	4.8	4.8	4.9	4.2	4.3	4.5
	5% WB	MCDBR	11.5	10.7	9.9	9.6	7.9	6.9	8.2	9.7	11.5	10.6	10.8	10.0
		MCWBR	4.4	5.0	3.9	4.7	4.1	3.9	4.5	4.7	4.9	4.6	4.4	4.4
Clear-Sky Solar Irradiance	taub	0.348	0.346	0.328	0.320	0.310	0.300	0.301	0.308	0.335	0.344	0.348	0.340	
	taud	2.522	2.549	2.589	2.597	2.611	2.624	2.611	2.572	2.484	2.462	2.482	2.531	
	Ebn at Noon	989	970	953	905	862	847	864	905	929	960	983	1001	
	Edh at Noon	111	105	94	83	72	67	71	83	102	113	115	111	
All-Sky Solar Radiation	RadAvg	7.88	6.98	5.70	4.23	2.99	2.48	2.70	3.67	4.98	6.30	7.31	7.76	
	RadStd	0.38	0.34	0.24	0.21	0.12	0.12	0.12	0.20	0.34	0.33	0.33	0.31	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-92	+157	+69	

Nomenclature: See separate page